Mathematics practice booklet: Grade 5 expectations

Grade 5 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 5 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: 5.OA.A.1, 5.OA.A.2, 5.OA.B.3

A worked example

Work out 4 x (6 + 3) - 5. Then write the expression for: add 6 and 3, then multiply the sum by 4.

1. Brackets first: 6 + 3 = 9.
2. Then multiply: 4 x 9 = 36.
3. Then subtract: 36 - 5 = 31.
4. Expression for the words: 4 x (6 + 3).

31; 4 x (6 + 3)

Questions

1. Evaluate: 18 - (4 + 5)

2. Evaluate: 3 x (12 - 7) + 8

3. Evaluate: {20 - [3 x (2 + 1)]} / 2

4. Write as an expression: three times the sum of 8 and 4.

5. Without calculating, is 2 x (45 + 17) bigger or smaller than 45 + 17?

6. Rule A: start at 0 and add 3. Rule B: start at 0 and add 6. Write the first five numbers of each.

7. How does each number in Rule B compare with the matching number in Rule A?

8. Write the ordered pairs from the last two questions (A first, B second).

---

9. A taxi charges AED 10 to start and AED 2 for each km. Write a rule for the cost and find the cost of 6 km.

---

10. Insert brackets to make this true: 2 + 3 x 4 = 20

Section 2: Shapes, measuring and space

Standards practised: 5.MD.A.1, 5.MD.C.3, 5.MD.C.4, 5.MD.C.5, 5.G.A.1, 5.G.A.2, 5.G.B.3, 5.G.B.4

A worked example

A fish tank is 50 cm long, 30 cm wide and 40 cm high. Find its volume in cubic centimetres and in litres (1 litre = 1,000 cubic cm).

1. Volume = length x width x height.
2. 50 x 30 = 1,500, then 1,500 x 40 = 60,000 cubic cm.
3. 60,000 / 1,000 = 60 litres.

60,000 cubic cm, which is 60 litres

Questions

1. A box is 5 cm by 4 cm by 3 cm. What is its volume?

2. A cube has edges of 4 cm. What is its volume?

3. A box has a volume of 120 cubic cm. Its base is 10 cm by 4 cm. How tall is it?

4. Convert 2.5 m to centimetres.

5. Convert 3,200 mL to litres.

6. Point A is at (2, 3) on a grid. Starting at the origin, describe how to get there.

7. Plot the points (1, 1), (4, 1) and (4, 3). What is the area of the triangle they make?

8. Is every square also a rectangle? Explain.

---

9. Is every rectangle also a square? Explain.

---

10. A pool is 10 m by 5 m with water 2 m deep. How many cubic metres of water fill it?

Section 3: Data, graphs and chance

Standards practised: 5.MD.B.2

A worked example

Water in 5 cups measures 1/2, 1/4, 1/4, 1/2 and 1/2 of a litre. If all the water is poured evenly into the 5 cups, how much does each hold?

1. Add: 1/2 + 1/4 + 1/4 + 1/2 + 1/2 = 2/4 + 1/4 + 1/4 + 2/4 + 2/4 = 8/4 = 2 litres.
2. Share equally between 5 cups: 2 / 5 = 2/5 of a litre.

2/5 of a litre

Questions

1. A line plot shows 4 dots at 1/4 and 2 dots at 1/2. What is the total?

2. Find the mean of 3, 5, 6, 10.

3. Find the median of 3, 5, 6, 10, 12.

4. Daily steps in thousands: 6, 8, 8, 9, 14. Which is more useful to describe a typical day: the mean or the median? Why?

5. The scores 4, 6, 6, 8 have the same mean and median. Is that always true?

6. A line plot of eggs per nest: 3 eggs (2 nests), 4 eggs (5 nests), 5 eggs (3 nests). How many nests?

7. Using the same data, how many eggs are there in all?

8. What is the mode in that data?

---

9. Why might a graph with a scale that starts at 50 instead of 0 be misleading?

---

10. Collect the heights of 6 people at home, in cm. Find the range.

Section 4: Numbers and calculating

Standards practised: 5.NBT.A.1, 5.NBT.A.2, 5.NBT.A.3, 5.NBT.A.4, 5.NBT.B.5, 5.NBT.B.6, 5.NBT.B.7, 5.NF.A.1, 5.NF.A.2, 5.NF.B.3, 5.NF.B.4, 5.NF.B.5, 5.NF.B.6, 5.NF.B.7

A worked example

What is 2/3 + 1/4? What is 2/3 x 3/5?

1. Unlike denominators: find a common one, 12. 2/3 = 8/12 and 1/4 = 3/12.
2. 8/12 + 3/12 = 11/12.
3. Multiplying: top times top and bottom times bottom: 2 x 3 = 6 and 3 x 5 = 15, giving 6/15 = 2/5.

11/12 and 2/5

Questions

1. What is 10 times 3.4? What is one tenth of 3.4?

2. Write 4.075 in words.

3. Compare using <, > or =: 0.62 and 0.6

4. Round 7.846 to the nearest hundredth.

5. Multiply: 346 x 27

6. Divide: 1,248 / 24

7. Add: 12.45 + 3.8

8. Subtract: 20 - 7.35

---

9. Calculate: 3/4 - 1/6

---

10. 4 friends share 3 pizzas equally. How much pizza does each get? Write it as a fraction.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 9
2. 23
3. 5.5 (inside: 2 + 1 = 3, 3 x 3 = 9, 20 - 9 = 11, 11 / 2)
4. 3 x (8 + 4)
5. Bigger: it is twice the sum.
6. A: 3, 6, 9, 12, 15. B: 6, 12, 18, 24, 30.
7. It is twice as big.
8. (3, 6), (6, 12), (9, 18), (12, 24), (15, 30)
9. Cost = 10 + 2 x km; 6 km costs AED 22.
10. (2 + 3) x 4 = 20

Section 2: Shapes, measuring and space

1. 60 cubic cm
2. 64 cubic cm
3. 3 cm
4. 250 cm
5. 3.2 litres
6. Go 2 units right, then 3 units up.
7. 3 square units (base 3, height 2, half of 6)
8. Yes: a square has four right angles and opposite sides equal.
9. No: a rectangle's sides need not all be equal.
10. 100 cubic metres

Section 3: Data, graphs and chance

1. 2 (4 x 1/4 = 1 and 2 x 1/2 = 1)
2. 6
3. 6
4. The median (8): the one very high day pulls the mean up.
5. No. For example 1, 2, 9: the mean is 4 and the median is 2.
6. 10
7. 41 (6 + 20 + 15)
8. 4 eggs
9. Small differences look much bigger than they are.
10. Answers vary: largest minus smallest.

Section 4: Numbers and calculating

1. 34 and 0.34
2. Four and seventy-five thousandths
3. 0.62 > 0.6
4. 7.85
5. 9,342

6. 52
7. 16.25
8. 12.65
9. 7/12 (9/12 - 2/12)
10. 3/4 of a pizza